

WE CREATE YOUR TOMORROW!



Stainless and Heat resistant super alloy

Hard-to-cut material cutting solution

- Turning
- Milling
- Endmill





We learn the challenge with nature.
We also dream love in nature.
Challenge for dream, Love for human
It's an ideal of Korloy like nature.

The 21st century of KORLOY is the power for a better future.

KORLOY Inc. is a total cutting tool manufacturer specializing in uncoated, coated (CVD, PVD), and Cermet inserts.

KORLOY Inc. was founded in 1966.

Our establishment is now a top leading cutting tool provider that constantly invests and improves our R&D center in Korea.

Our goal is to be the top, globally leading company that overcomes the continuous challenges and innovations.

KORLOY Inc. products manufactured by the ever growing R&D department along with our innovative production technology are selling and exporting to over 80 countries, as the global company recognized with higher technology.

KORLOY Inc. continuously endeavors to grow as the global enterprise representing Korea.

We sincerely anticipate your on-going interest for KORLOY Inc.



C O N T E N T S



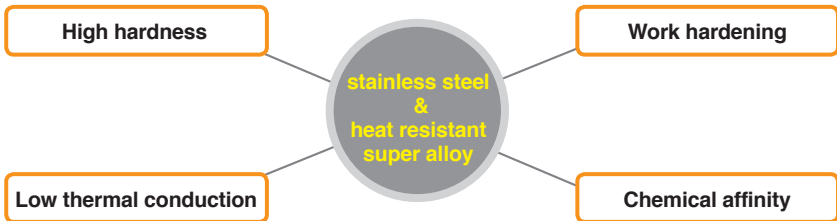
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Hard-to-cut material cutting

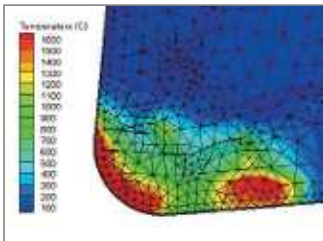
General trouble cases of hard-to-cut material such as stainless steel and heat resistant super alloy are wear of top and side cutting edge, notch wear and built-up edge by adhesion.

We, Korloy Inc, would like to suggest new grade line-up for better performance in stainless steel and heat resistant super alloy.

Factor of low tool life



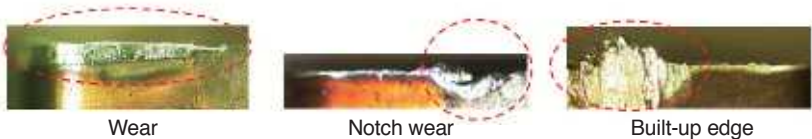
Thermal analysis for cutting edge (Inconel 718)



Generally, 800~1000°C high temperature is generated in small area during stainless steel and heat resistant super alloy cutting. Moreover, cooling by chip evacuation is not so effective due to low thermal conduction.

Therefore, wear and breakage of the tool are easily made.

Edge failure case



Wear

Notch wear

Built-up edge

Grade select guide

Grade	Coating	ISO	Maching type	Application	Turning	Milling	Face turning	Grooving	Threading	Parting off	Drill	Endmill
New NC9025	CVD	M25~M35 S15~S25	●	Stainless steel and heat resistant super alloy	●							
New NC5330	CVD	P30~P40 K15~K25 M25~M35 S15~S25	●	Stainless steel and heat resistant super alloy, mild steel, forged steel, general steel	●	●	●	●		●	●	
New PC8110	PVD	M10~M20 S10~S20	●	Stainless steel and heat resistant super alloy, mild steel, forged steel, general steel	●			●		●		
New PC5330	PVD	P35~P45 K10~K20 M20~M30 S20~S30	●	Stainless steel and heat resistant super alloy, cast iron, general steel	●	●	●	●	●	●	●	●
PC9030	PVD	M30~M40	●	Stainless steel	●		●	●		●		
PC9530	PVD	M30~M40	●	Stainless steel		●					●	
PC3030T	PVD	M25~M35	●	Stainless steel					●			
PC3545	PVD	P40~P50 M40~M50 S30~S40	●	Heat resistant super alloy, general steel		●						
New PC210	PVD	S10~S25	●	Stainless steel and heat resistant super alloy								●

Maching type : ● Continuous cutting
 ● General cutting
 ● Interrupted cutting

Hard-to-cut material
cutting

New grade for Hard-
to-cut material

Technical guide
for Turning

Technical guide
for Milling

Technical guide
for Endmill

Workpiece
code table

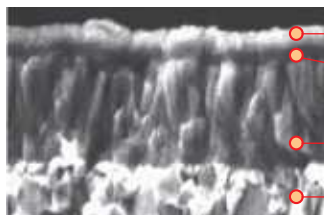
Hard-to-cut material
cutting solution

Trouble shooting

New grade for hard-to-cut material cutting

NC9025/NC5330(CVD)

Coating layer



TiN : Good anti adhesion & surface roughness

Al₂O₃ : Anti oxidation

MT CVD TiCN : Fine grain & columnar structure with high toughness and hardness

Exclusive substrate for coating layer

Surface of coating layer / Substrate



Conventional

Less B.U.E.

Smooth surface
(Good surface roughness)



New NC9025 / NC5330



Conventional

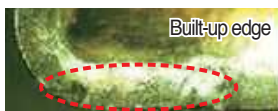
Less notch wear

Tough substrate



New NC9025 / NC5330

Performance



Built-up edge

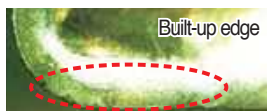


Anti chipping Notch wear

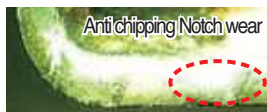


Conventional

Less B.U.E.
Less chipping



Built-up edge



Anti chipping Notch wear



New NC9025

■ Cutting condition

- vc=200(m/min)
- fn=0.25(mm/rev)
- ap=1.5(mm)
- wet
- Wear resistance test

■ Test Tool

- Holder
DCLNR2525-M12
- Insert
CNMG120408-HS

PC8110/PC5300(PVD)

Coating layer



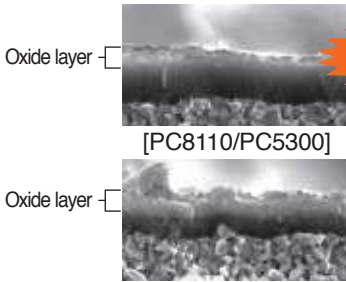
Higher hardness → better wear resistance
Anti oxidation → longer tool life

Good bonding strength of coating and substrate
Good anti chipping

Fine grain tough substrate

Anti-oxidation

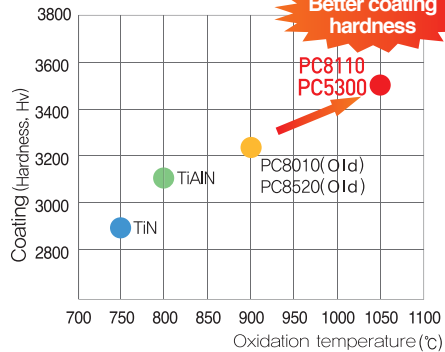
•1 hour heating under air at 900 °C



Better anti oxidation

[Competitor]

Coating hardness at high temperature



Better coating hardness

Surface of coating layer / Substrate

Smooth surface
(Good surface roughness)

Coating surface

Good surface roughness

Coating surface

Tough substrate
(Anti chipping)

Substrate

Tougher substrate

Substrate

[PC5300]

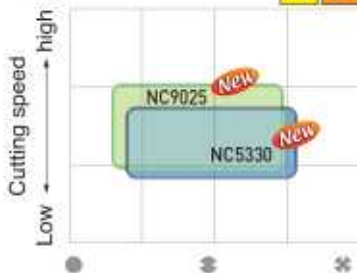
[Competitor]

Technical guide for Turning

Turning grade choice

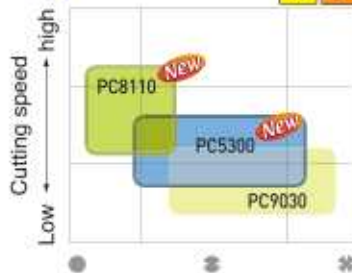
CVD

M S



PVD

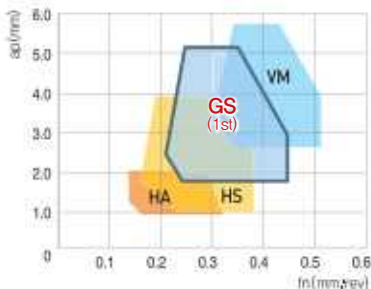
M S



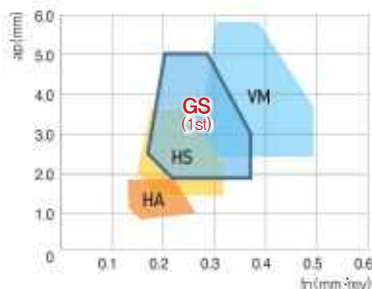
* See page 33 : CVD & PVD grade choice for trouble shooting

Turning chip breaker choice

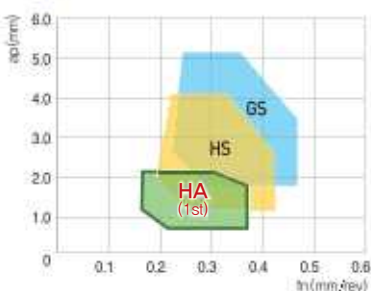
Stainless - Negative



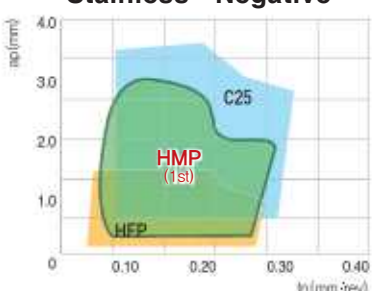
Inconel - Negative



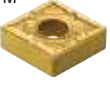
Titanium alloy - Negative



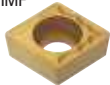
Inconel, Titanium alloy, Stainless - Negative



Features of chip breaker (negative)

Geometry	Cross section view	Depth of cut(mm)	Feed(mm/rev)	Features
HA 		0.5 ~ 1.5 ~ 2.5 Medium - finish	0.03 ~ 0.15 ~ 0.30	• Sharp edge : Low cutting force • Good chip control at high speed • High rake (suitable for Ti-alloy)
HS 		1.0 ~ 2.5 ~ 4.0 Medium	0.1 ~ 0.25 ~ 0.40	• Good chip control • High rake with strong edge • Tougher geometry to prevent notch wear
GS 		1.5 ~ 3.0 ~ 5.5 Continuous Medium - roughing	0.15 ~ 0.3 ~ 0.50	• Tough edge for medium intermittence • Low cutting force for less B.U.E. • Wide chip pocket for better chip flow
VM 		2.0 ~ 4.5 ~ 6.5 Intermittent Medium - roughing	0.20 ~ 0.40 ~ 0.60	• Tough edge for heavy intermittence • Strong cutting edge geometry

Features of chip breaker (positive)

Geometry	Cross section view	Depth of cut(mm)	Feed(mm/rev)	Features
HFP 		0.1 ~ 0.5 ~ 1.5 Finish	0.05 ~ 0.15 ~ 0.25	• Good chip control at shallow depth • Good surface and low cutting force
HMP 		0.5 ~ 1.5 ~ 3.0 Medium - finish	0.10 ~ 0.20 ~ 0.30	• Good chip control at wide feed and depth • General use
C25 		1.0 ~ 1.5 ~ 3.0 Medium	0.15 ~ 0.25 ~ 0.35	• Suitable for intermittence • Strong cutting edge

Recommended cutting speed

ISO	Workpiece				Cutting speed v_c (m/min) $f_n \sim 0.3$ mm/rev, $a_p \sim 4$ mm			
	Kind	ISO	DIN	Hardness (HB)	New CVD NC9025 NC5330	New PVD PC8110	New PVD PC5300	PVD PC9030
M	Ferrite Stainless Steel	X6CrAl13	—	165	90~230	150~250	100~220	80~170
		X6Cr17	X6CrMo17-1	180	90~230	150~250	100~220	80~170
		X2CrMoTi18-2	—	220	80~220	120~230	90~200	70~150
	Austenitic Stainless Steel	X5CrNi18-9	X2CrNi19-11	180	80~250	150~280	100~240	70~190
		X5CrNiMo17-12-2	X2CrNiMo17-13-2	200	80~220	120~250	90~210	70~160
	Martensite Stainless Steel	X12Cr13	X20Cr13	200	90~230	130~250	100~220	80~180
		X20Cr13	—	220	90~230	130~250	100~220	80~180
		X70CrMo15	X7CrNiAl17-7	230	80~220	130~240	100~210	70~180
	Precipitation hardening Stainless Steel	X5CrNiCuNb16-4	—	> 300	40~110	80~180	50~150	30~100
		X7CrNiAl17-7	—	> 300	40~110	80~180	50~150	30~100
S	Nickel base alloys	Inconel	—	220~300	15~50	40~90	20~70	15~50
		Nimonic	—	350	15~50	40~70	20~60	15~50
		Hastelloy	—	160	15~50	40~90	20~70	15~50
	Cobalt base alloys	Stellite	—	370	15~50	30~80	20~60	15~40
	Iron base alloys	17-4-PH	—	250	35~80	30~70	20~60	20~50
		Incoloy	—	180	35~80	30~70	20~60	20~50
	Titanium alloys	Ti-6246	TiAl5Sn2.5	200	40~150	60~200	50~170	40~120
		Ti-6Al-4V	TiAl6V4	250~310	40~150	50~180	40~150	40~100
M+S	Stainless steel + Inconel	—	—	—	15~50	40~80	20~60	15~50
	Stainless steel + Stellite	—	—	—	15~50	40~80	20~60	15~50



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Hard-to-cut material
cutting solution

Trouble shooting

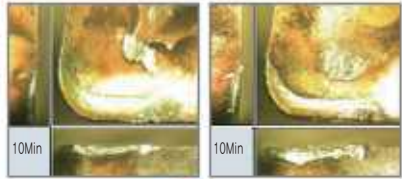
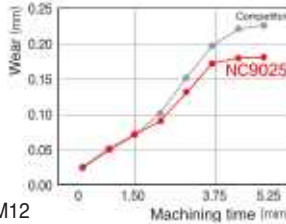
NC9025 turning test examples

M Stainless steel (X5CrNiMo17-12-2)

■ Cutting condition

- $vc=200(m/min)$
- $fn=0.25(mm/rev)$
- $ap=1.5(mm)$
- wet

■ Result



NC9025 **New**

Competitor (M30)

■ Test Tool

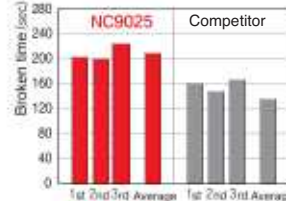
- Holder DCLNR2525-M12
- Insert CNMG120408-HS

M Stainless steel (X5CrNiMo17-12-2-Hole)

■ Cutting condition

- $vc=200(m/min)$
- $fn=0.25(mm/rev)$
- $ap=1.5(mm)$
- wet

■ Result



NC9025 **New**

Competitor (M30)

■ Test Tool

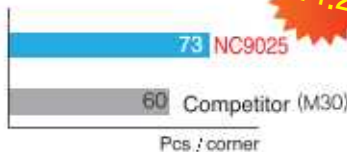
- Holder DCLNR2525-M12
- Insert CNMG120408-HS

M Stainless steel (X5CrNi18-9)

■ Cutting condition

- $vc=197(m/min)$
- $fn=0.30(mm/rev)$
- $ap=0.5(mm)$
- dry

■ Result



■ Test Tool

- Holder DTGMR2525-M16
- Insert TNMG160408-HS

NC9025 turning test examples

S Stellite

■ Cutting condition

- $vc=32(m/min)$
- $fn=0.14(mm/rev)$
- $ap=5.0(mm)$
- wet

■ Test Tool

- Holder DCLNR2525-M19
- Insert CNMG190612-GS

■ Result



M Stainless steel (X5CrNi18-9)

■ Cutting condition

- $vc=150(m/min)$
- $fn=0.30(mm/rev)$
- $ap=6.0(mm)$
- wet

■ Test Tool

- Holder DCLNR2525-M19
- Insert CNMG190608-B25

■ Result



M Stainless steel (X5CrNi18-9)

■ Cutting condition

- $vc=115(m/min)$
- $fn=0.20(mm/rev)$
- $ap=2.0(mm)$
- wet

■ Test Tool

- Holder DCLNR2525-M12
- Insert CNMG120408-HS

■ Result





Hard-to-cut material
cutting

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Workpiece
code table

Hard-to-cut material
cutting solution

Trouble shooting

PC8110 turning test examples

S Inconel718

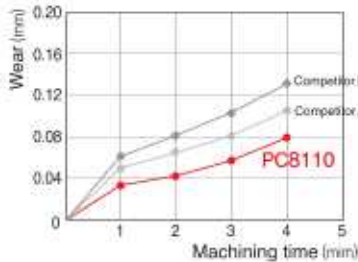
■ Cutting condition

- $vc=60(m/min)$
- $fn=0.2(mm/rev)$
- $ap=2.0(mm)$
- wet
- 4min

■ Test Tool

- Holder DCLNR2525-M12
- Insert CNMG120408-GS

■ Result



S Titanium alloy (Ti-6Al-4V)

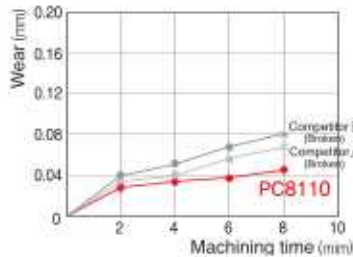
■ Cutting condition

- $vc=70(m/min)$
- $fn=0.2(mm/rev)$
- $ap=1.0(mm)$
- wet
- 8min

■ Test Tool

- Holder DCLNR2525-M12
- Insert CNMG120408-HA

■ Result



S M Stainless steel + stellite

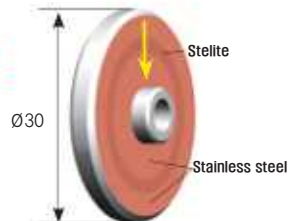
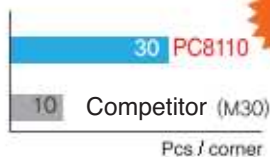
■ Cutting condition

- $vc=60(m/min)$
- $fn=0.2(mm/rev)$
- $ap=2.0(mm)$
- wet

■ Test Tool

- Holder DCLNR2525-M12
- Insert CNMG120408-GS

■ Result



PC8110, PC5300 turning test examples

M Stainless steel (X5CrNiMo17-12-2)

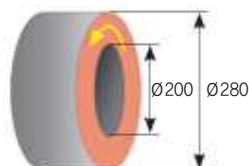
■ Cutting condition

- $vc=200(\text{m/min})$
- $fz=0.2(\text{mm/rev})$
- $ap=2.0(\text{mm})$
- wet

■ Test Tool

- Holder DCLNR2525-M12
- Insert CNMG120408-GS

■ Result



M Stainless steel (X5CrNi18-9)

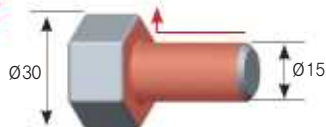
■ Cutting condition

- $vc=282(\text{m/min})$
- $fn=0.2(\text{mm/rev})$
- $ap=3(\text{mm})$

■ Test Tool

- wet
- Test Tool
- Holder DCLNR2525-M12
- Insert CNMG120408-HS

■ Result



M Stainless steel (X5CrNiMo17-12-2)

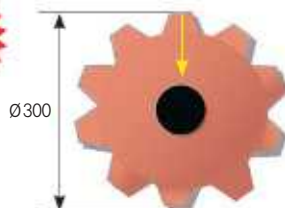
■ Cutting condition

- $vc=120(\text{m/min})$
- $fn=0.2(\text{mm/rev})$
- $ap=0.5\sim 1.5(\text{mm})$
- wet

■ Test Tool

- Holder DSBNR2525-M12
- Insert SNMG120408-GS

■ Result





Stock management items (Turning insert)

NC9025(CVD)

CNMG 090304-HS	CNMG 190616-HS	SNMG 090304-HS	TNMG 160404-HS	WNMG 060404-HA
090308-HS	DNMG 110404-HS	120404-GS	160408-GS	060408-GS
120404-GS	110408-HS	120404-HA	160408-HA	060408-HA
120404-HA	150404-HA	120404-HS	160408-HS	060408-HS
120404-HS	150604-HS	120408-GS	160412-GS	060412-HS
120408-GS	150408-HA	120408-HA	160412-HS	080404-GS
120408-HA	150408-HS	120408-HS	220408-GS	080404-HA
120408-HS	150604-GS	120412-GS	220408-HA	080404-HS
120412-GS	150604-HA	120412-HS	220408-HS	080408-GS
120412-HA	150604-HS	150612-HS	220412-HS	080408-HA
120412-HS	150608-HA	190612-GS	VNMG 160404-HS	080408-HS
190612-GS	150608-HS	190612-HS	160408-HS	080412-GS
160612-HS	150612-GS	190616-HS	WNMG 060404-GS	080412-HS
190612-HS	150612-HS	TNMG 160404-GS	060404-HA	

* For more information, please refer KORLOY 2010-2011 catalogue

NC5330(CVD)

CNMG 090304-VF	CNMG 160608-B25	DNMG 150408-HC	090308-VF	SNMM 190616-GH
090304-VM	160612-GR	150408-VM	090308-VM	190624-GH
090308-VF	190604-B25	150412-B25	120404-B25	250724-GH
090308-VM	190608-B25	150412-VM	120404-VF	250724-VT
120404-B25	190608-GR	150604-B25	120404-VM	250924-GH
120404-HC	190612-B25	150604-HC	SNMG 120408-B25	TNMG 110304-VF
120404-VF	190612-GR	150604-VF	120408-GR	160404-B25
120404-VL	190616-B25	150604-VM	120408-VM	160404-HC
120404-VW	190616-GR	150608-B25	120408-VF	160404-VF
120408-B25	250924-GR	150608-GR	120412-B25	160404-VM
120408-GM	CNMM 120408-GH	150608-VF	120412-VM	160408-B25
120408-GR	120412-GH	150608-VM	120416-B25	160408-HC
120408-HC	190608-GH	150612-B25	150608-B25	160408-VF
120408-HR	190612-GH	150612-VM	190608-B25	160408-VM
120408-LW	190612-VT	150616-GR	190612-B25	160412-HC
120408-VF	190616-GH	KNUX 160405-L11	190612-GR	160412-VM
120408-VL	190624-GH	160405-R11	190616-B25	220404-B25
120408-VM	250924-GH	160405-R12	190616-GR	220404-VF
120412-B25	250724-VT	160410-L11	250716-B25	220408-B25
120412-GR	DNMG 110404-VF	160410-L12	250724-B25	220408-GR
120412-HC	110408-VF	160410-R11	SNMM 120408-GH	220408-VM
120412-HR	110408-VM	160410-R12	120412-GH	220412-VM
120412-VM	150404-B25	RNMG 120400-B25	150612-GH	TNMX 160404R
120416-VM	150404-HC	SNMG 090304-VF	190612-GH	220404R
	150404-VM	090308-B25	190612-VT	VNMG 160404-VF

* For more information, please refer KORLOY 2010-2011 catalogue

NC5330(CVD)

CCMT	060204-C25	DCMT	070204-HMP	MRMN	400-M	TCMT	110204-C25	WNMG	060404-VF
	060204-HMP		070208-HMP		500-M		110204-HFP		060404-VM
	060208-C25		11T302-HFP		600-M		110204-HMP		060408-VF
	060208-HMP		11T302-HMP		800-M		110208-C25		060408-VM
	09T302-HFP		11T304-C25	RCMX	1003M0		110208-HMP		060412-LW
	09T302-HMP		11T304-HMP		1204M0		16T304-C25		080404-VF
	09T304-C25		11T308-C25		1606M0		16T304-HMP		080404-VM
	09T304-HFP		11T308-HMP	SCMT	09T304-C25		16T308-C25		080408 GR
	09T304-HMP		MFMN 300		09T308-C25		16T308-HMP		080408-HC
	09T308-C25		MGML 300-6D-PT		09T308-HMP	TPMR	160304-F		080408-HR
	09T308-HMP		MGMN 200-G		120404-C25		TPMT 110304-VF		080408-LW
	120404-C25		200-M	SP	200		110308-VF		080408-VF
	120404-HMP		300-G		300	VBMT	160404-HMP		080408-VM
	120408-C25		300-02-M		400		160408-HMP		080412 GR
	120408-HMP		400-M		500	VCMT	080204-VF		080412-LW
CPMT	090304-VF	DCMT	500-M	TCMT	600		VNMG 160404-VM		080412-VM
	090308-VF		600-M		090204-C25		160408-VF		
	070202HFP		800-M		090208-C25		160408-VL		
DCMT	070204-C25	MRMN	300-M		090208-HMP		160408-VM		

* For more information, please refer KORLOY 2010-2011 catalogue

PC8110(PVD)

CCMT	060202-HMP	CNMG	190616-GS	FMM	400R-04	SP	400L	VNMG	160404-HS	
	060204-HMP		190616-HS		500R-04		400R		160408-HA	
	060208-HMP	DCMT	070204-HMP	SNEU	120420-MF	500	160408-HS			
	09T302-HMP		11T302-HMP	SNMG	120408-GS	TCMT	16T304-HMP	WNMG	060404-GS	
	09T304-HMP		11T304-HMP		120408-HA		TNMG	160404-GS	060404-HA	
	09T308-HMP		11T308-HMP		120408-HS		160404-HA		060404-HS	
CNMG	120404-GS	DNMG	150404-GS		120408-VM		160404-HS		060408-GS	
	120404-HA		150404-HA		120412-GS		160408-GS	060408-HA		
	120404-HS		150404-HS		120412-HS		160408-HA	060408-HS		
	120408-GS		150408-GS		150612-HS		160408-HS	080404-GS		
	120408-HA		150408-HA		190612-GS		220408-GS	080404-HA		
	120408-HS		150408-HS		190612-HS		220408-HS	080404-HS		
	120408-VM		150604-GS		190616-HS		220408-VM	080408-GS		
	120412-GS		150604-HA		SP		200	TPMR	160304-F	080408-HA
	120412-HS		150604-HS				200R		160308-M	080408-HS
	120412-VM		150608-GS			300	VBMT		160404-HMP	080408-VM
	160612-VM		150608-HA		300L	160408-HMP				
	190612-GS		150608-HS		300R	VNGG	160408-HA			
	190612-HS		FMM		300R-03	400	VNMG	160404-HA		

* For more information, please refer KORLOY 2010-2011 catalogue



PC5300(PVD)

CCGT	060204-HFP	CNMG	120412-HS	DNMG	150608-GS	SNMG	190616-HS	VNMG	160408-HS
CCMT	060204-HMP		120412-GS	MGMN	150-G		120408-GS	WNMG	060404-HS
	09T304-HMP	DCGT	11T304-HFP		250-G	TCMT	110204-HMP		060408-GS
	09T308-HMP		11T308-HFP		200-M		110208-HMP		060408-HS
	120404-HMP	DCMT	11T304-HMP		250-M		16T304-HMP		060412-HS
	120408-HMP		11T308-HMP		300-M		16T308-HMP		080404-GS
CNMG	120404-B25	DNMG	150408-HA		400-G	TNMG	160404-HS		080408-GS
	120404-HA		150404-HS	SCMT	09T304-HMP		160408-GS		080408-HS
	120404-HS		150408-HS		09T308-HMP		160408-HS		
	120404-GS		150412-HS	SNMG	120404-HS		160412-B25		
	120408-B25		150604-HS		120408-B25		160412-HS		
	120408-HA		150608-HS		120408-HS		220404-B25		
	120408-HS		150408-GS		120412-HS		220408-HS		
	120408-GS		150604-GS		190612-HS	VBMT	160408-HMP		

* For more information, please refer KORLOY 2010-2011 catalogue

PC9030(PVD)

CCET	030102L	CNMG	120404-HA	DCMT	070204-C25	DNMG	150612-HS	SCR	081020-R05
	040102L		120404-HM		070204-HMP	KNUX	160405R-11		081020-R10
CCGT	060201R-KF		120404-HS		11T302-HMP	MGMN	200-G		081025-R05
	060201R-KM		120404-VM		11T304-C25		200-M		081025-R10
	060202R-KF		120408-B25		11T304-HMP		250-G		081030-R05
	060202R-KM		120408-GS		11T308-C25		250-M		081030-R10
	09T301R-KF		120408-HA		11T308-HMP		300-M	SGR	060610-R05
	09T301R-KM		120408-HM	DNMG	110404-HS	SBR	060520-10-R05		060610-R10
	09T302R-KF		120408-HS		110408-HS		060520-10-R10		060615-R05
	09T302R-KM		120408-VM		150404-HA		060630-20-R05		060615-R10
	09T304-HFP		120412-GS		150404-HS		060630-20-R10		060620-R05
CCMT	060202-HMP		120412-HA		150408-HA		080630-20-R05		060620-R10
	060204-C25		120412-HS		150408-HS		080630-20-R10		081015-R05
	060204-HMP		160612-HS		150412-HS		080840-20-R05		081015-R10
	060208-HMP		160616-HS		150604-GS		080840-20-R10		081020-R05
	09T302-HMP		190612-HS		150604-HA	SCMT	120404-C25		081020-R10
	09T304-C25		190616-HS		150604-HM		120408-C25		081025-R05
	09T304-HMP	DCGT	070201R-KF		150604-HS	SCR	060610-R05		081025-R10
	09T308-C25		070201R-KM		150604-VM		060610-R10		081030-R05
	09T308-HMP		070202R-KF		150608-B25		060615-R05		081030-R10
	120404-HMP		070202R-KM		150608-GS		060615-R10	SNMG	090304-HS
	120408-HMP		11T301R-KF		150608-HA		060620-R05		120404-HA
CNMG	090304-HS		11T301R-KM		150608-HM		060620-R10		120404-HM
	090308-HS		11T302R-KF		150608-HS		081015-R05		120404-HS
	120404-GS		11T302R-KM		150608-VM		081015-R10		120408-B25

* For more information, please refer KORLOY 2010-2011 catalogue

PC9030(PVD)

SNMG 120408-GS	SP 400	TNMG 160408-HS	VBMT 160404	WNMG 060404-HS
120408-HA	TBGT 060102L	160408-VM	160408	060408-HA
120408-HM	TCMT 110204-HMP	160412-HS	160408-HMP	060408-HS
120408-HS	110208-C25	220408-HA	VCMT 110301R-KF	060412-HS
120408-VM	16T304-C25	220408-HS	110301R-KM	080404-GS
120412-HM	16T304-HMP	220412-HS	110302R-KF	080404-HA
120412-HS	16T308-C25	TPGH 080202L	110302R-KM	080404-HS
150612-HS	16T308-HMP	110202L	VNMG 160404-HM	080408-GS
150616-HS	TNMG 160404-GS	110204L	160404-HS	080408-HA
190612-HS	160404-HA	TPMR 110304-F	160408-HM	080408-HM
190616-HS	160404-HS	160304-F	160408-HS	080408-HS
SP 200	160408-B25	VBGT 110301R-KF	160408-VM	080408-VM
200L	160408-GS	110301R-KM	WBGT 020102L	080412-HS
200R	160408-HA	110302R-KF	S30202L	
300	160408-HM	110302R-KM	WNMG 060404-HA	

* For more information, please refer KORLOY 2010-2011 catalogue

PC3030T(PVD)

EL 16-1.0ISO	ER 16-10APIRD	ER 16-2.0ISO	ER 16-AG55	ER 27-4API382
16-1.5ISO	16-10UN	16-2.5ISO	16-AG60	27-4API383
16-2.0ISO	16-10W	16-20UN	16-G55	27-4API502
16-2.5ISO	16-11.5NPT	16-20W	16-G60	27-4API503
16-3.0ISO	16-11BSPT	16-24UN	22-3.0SAGE	27-6.0TR
16-A55	16-11UN	16-24W	22-3.5ISO	27-Q55
16-A60	16-11W	16-26W	22-4.0ISO	27-Q60
16-AG55	16-12UN	16-28UN	22-4.0TR	M16-1.5ISO
16-AG60	16-12UNJ	16-28W	22-4.5ISO	M16-1.75ISO
16-G60	16-12W	16-3.0ISO	22-4API382	M16-2.0ISO
22-5ACME	16-13UN	16-3.0TR	22-4RD	M16-3.0ISO
22-6ACME	16-14BSPT	16-32UN	22-5.0ISO	M16-A55
ER 11-24UNJ	16-14NPT	16-6RD	22-5.0TR	M16-A60
11-A60	16-14UN	16-8APIRD	22-5ACME	M16-AG55
16-0.4ISO	16-14W	16-8BBUT	22-5API403	M16-AG60
16-0.5ISO	16-16UN	16-8NPT	22-5BUT1	M16-G55
16-0.75ISO	16-16W	16-8RD	22-5BUT75	M16-G60
16-0.7ISO	16-18NPT	16-8UN	22-6ACME	M22-N55
16-0.8ISO	16-18NPTF	16-8W	22-6UN	M22-N60
16-1.0ISO	16-18UN	16-9UN	22-6W	IL 11-11UN
16-1.25ISO	16-18W	16-9W	22-N55	11-14NPT
16-1.5ISO	16-19BSPT	16-A55	22-N60	16-11.5NPT
16-1.75ISO	16-19W	16-A60	27-4.0SAGE	16-2.5ISO

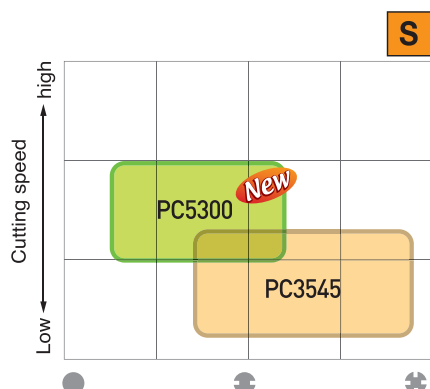
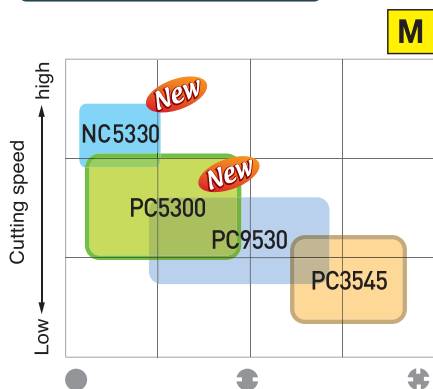
* For more information, please refer KORLOY 2010-2011 catalogue

PC3030T(PVD)

IL 16-8NPT	IR 16-0.7ISO	IR 16-18NPT	IR 16-AG60	IR 27-4W
16-A60	16-0.8ISO	16-18UN	16-G55	27-5.5ISO
16-AG60	16-1.0ISO	16-18W	16-G60	27-Q55
IR 11-0.5ISO	16-1.25ISO	16-19BSPT	22-3.5ISO	27-Q60
11-0.75ISO	16-1.5ISO	16-19W	22-4.0ISO	IRM 11-1.5ISO
11-1.0ISO	16-1.75ISO	16-2.0ISO	22-4.0TR	11-A55
11-1.25ISO	16-10ABUT	16-2.5ISO	22-4.5ISO	11-A60
11-1.5ISO	16-10APD	16-20UN	22-5.0ISO	16-1.5ISO
11-1.75ISO	16-10UN	16-20W	22-5.0TR	16-2.0ISO
11-11UN	16-10W	16-24UN	22-5ACME	16-3.0ISO
11-14NPT	16-11.5NPT	16-26W	22-5API403	16-A55
11-14W	16-11BSPT	16-28UN	22-5BUT1	16-A60
11-16W	16-11UN	16-3.0ISO	22-5BUT75	16-AG55
11-18NPT	16-11W	16-3.0TR	22-6API551	16-AG60
11-19W	16-12UN	16-8APD	22-N55	16-G55
11-2.0ISO	16-12W	16-8NPT	22-N60	16-G60
11-2.5ISO	16-14BSPT	16-8UN	27-4.0SAGE	22-N55
11-27NPT	16-14NPT	16-9UN	27-4.5W	22-N60
11-A55	16-14UN	16-9W	27-4API382	
11-A60	16-14W	16-A55	27-4API383	
16-0.5ISO	16-16UN	16-A60	27-4API502	
16-0.75ISO	16-16W	16-AG55	27-4API503	



Milling grade choice



Recommended cutting speed

ISO	Workpiece				Cutting speed v_c (m/min) $f_z \sim 0.3\text{mm/rev}$, $a_p \sim 4\text{mm}$			
	Kind	ISO	DIN	Hardness	PC5300 ^{New}	PC9530 ^{New}	PC3545	NC5330 ^{New}
M	Ferrite Stainless Steel	X6Cr17	X6CrMo17-1	165	100~180	70~150	70~120	120~200
	Austenitic Stainless Steel	STS304	X2CrNi19-11	180	120~220	100~200	80~180	140~240
	Martensite Stainless Steel	X12Cr13	X20Cr13	200	80~150	70~130	60~110	100~170
	Precipitation hardening Stainless Steel	STS630	—	300~	80~120	60~110	50~100	100~140
S	Nickel base alloys	X5CrNiCaNb16-4	—	220~300	20~70	—	20~50	—
	Cobalt base alloys	Stellite	—	370	20~60	—	20~50	—
	Iron base alloys	17-4-PH	—	250	20~70	—	20~50	—
	Titanium alloys	Ti-6246	TiAl6v4	200	70~150	—	50~120	—
M+S	Stainless steel + Inconel	—	—	—	20~70	—	20~50	—
	Stainless steel + Stellite	—	—	—	20~70	—	20~50	—



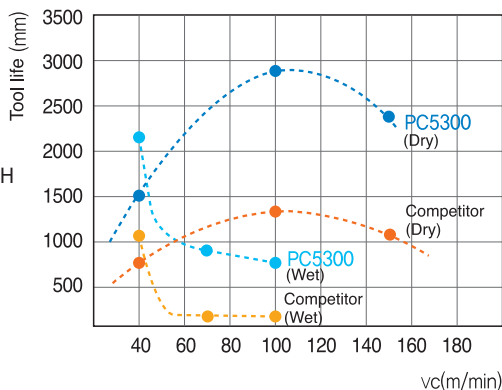
PC5300 test examples

Tool life as per cutting speed

M Stainless steel (X5CrNi18-9)

- Cutting condition
 - $fz=0.21$ (mm/t)
 - $ap=0.5\sim1.5$ (mm)
 - wet

- Test Tool
 - Cutter EPN4125R
 - Insert SPCN1203EDER-RH

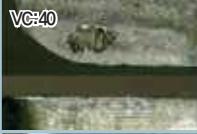


Result

Dry cutting

Cutting length	750mm	1300mm	1100mm
PC5300			
Competitor			

Wet cutting

Cutting length	750mm	1300mm	1100mm
PC5300			
Comptitor			

M Stainless steel (X5CrNi18-9)

- **Cutting condition**
- $vc=200(\text{m/min})$
 - $fz=0.15(\text{mm/t})$
 - $ap=2.0(\text{mm})$
 - dry

■ **Test Tool**

- Cutter EPN5160R
- Insert SPKN1504AESN-SU

■ **Result****M Stainless steel (X5CrNi18-9)**

- **Cutting condition**
- $vc=175(\text{m/min})$
 - $fz=0.18(\text{mm/t})$
 - $ap=4.0(\text{mm})$
 - dry

■ **Test Tool**

- Cutter EPN4125R
- Insert SPKN1203EDER-RH

■ **Result****M Stainless steel (X5CrNi18-9)**

- **Cutting condition**
- $vc=120(\text{m/min})$
 - $fz=1.2(\text{mm/t})$
 - $ap=1.2(\text{mm})$
 - dry

■ **Test Tool**

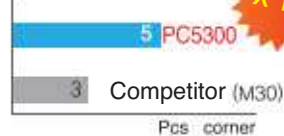
- Cutter HRMDCM13050H12-3
- Insert WNMX130520ZNN-MM

■ **Result****M Stainless steel (X5CrNiMo17-12-2)**

- **Cutting condition**
- $vc=65(\text{m/min})$
 - $fz=0.14(\text{mm/t})$
 - $ap=3.0(\text{mm})$
 - wet

■ **Test Tool**

- Cutter FMACM4100HR
- Insert SEET14M4AGSN-MM

■ **Result****M Stainless steel (X5CrNiMo17-12-2)**

- **Cutting condition**
- $vc=120(\text{m/min})$
 - $fz=0.2(\text{mm/t})$
 - $ap=2.0(\text{mm})$
 - dry

■ **Test Tool**

- Cutter EPN5160R
- Insert SPKN1504EDSR-SU

■ **Result**



Stock management items (Milling insert)

PC5300(PVD)

Geometry	Designation	Geometry	Designation
	APKT 1604PDSR-MF		NPMT 222408-DM
	1604PDSR-MM		222408-DS
	APMT 060202PDSR-MM		252808-DM
	060208PDSR-MM		252808-DS
	060212R-MM		293208-DM
	060216R-MM		293208-DS
	0602PDSR-MM		334008-DM
	090308PDSR-MM		334008-DS
	090312R-MM		415008-DM
	090316R-MM		415008-DS
	090320R-MM		516012-DM
	0903PDSR-MM		516012-DS
	11T308PDSR-MM		ONHX 0606ANN-MF
	11T312PDSR-MM		0606ANN-MM
	11T316R-MM		080608-W
	11T324R-MM		0806ANN-MF
	11T3PDSR-MF		0806ANN-MM
	11T3PDSR-MM		ONMX 0606ANN-MF
	160410PDSR-MM		0606ANN-MM
	160416PDSR-MM		0806ANN-MF
	160424R-MM		0806ANN-MM
	160432R-MM		RDKT 10T3M0-MF
	1604PDSR-MF		10T3M0-MM
	1604PDSR-MM		1204M0-MF
	180612PDSR-MM		1204M0-MM
	180616PDSR-MM		SDXT 09M405R-MF
	180620PDSR-MM		09M405R-MM
	180624PDSR-MM		130508R-MF
	1806PDSR-MF		130508R-MM
	1806PDSR-MM		SDCN 1504AEEN-RH
	LNEX 100605PNR-MF		SECN 1203AFEN-RH
	100605PNR-MM		SPCN 1203EDER-RH
	151008PNR-MF		1504EDER-RH
	151008PNR-MM		SEET 0903AGSN-MF
	LNMX 100605PNR-MF		0903AGSN-MM
	100605PNR-MM		14M4AGSN-MF
	151008PNR-MF		14M4AGSN-MM
	151008PNR-MM		14M4AGTN-MF
	LPMT 040203-DF		SEXT 0903AGSN-MF
			0903AGSN-MM
			14M4AGSN-MF
			14M4AGSN-MM
			14M4AGSN-MR

* For more information, please refer KORLOY 2010-2011 catalogue

PC5300(PVD)

Geometry	Designation		Geometry	Designation	
	SNEX	1206ANN-MF		SPMT	070204-DM
		1206ANN-MM			070204-DS
		1206ENN-MF			
		1206ENN-MM			
	SNMX	1206ANN-MF		TPCN	1603PDER-RH
1206ANN-MM		2204PDER-RH			
1206ENN-MF		TPKN		2204PDSR-SU	
1206ENN-MM					
	SPKN		WCMT	040208-C20	
				050308-C20	
				06T308-C20	
				080408-C20	
	SPMT	050203-DF		WDKT	080316ZDSR-MH
		050203-DM			10T320ZDSR-MH
		050203-DS			130520ZDSR-MH
		060204-DF			150625ZDSR-MH
		060204-DM		WNMX	09T316ZNN-MM
	060204-DS	09T316ZNN-MR			
	070204-DF	130520ZNN-MM			
		130520ZNN-MR			

* For more information, please refer KORLOY 2010-2011 catalogue

PC9530(PVD)

Geometry	Designation		Geometry	Designation	
	ADKT	1505PDSR-MM		NPMT	222408-DS
					252808-DS
					293208-DS
	APKT	1604PDSR-MM			
	APMT	0602PDSR-MM			415008-DS
		0903PDSR-MM			516012-DS
		1806PDSR-MM		OFKR	0704SN-MM
	APXT	1035PDSR-MM			
		11T3PDSR-MF		ONHX	0606ANN-MF
	11T3PDSR-MM	0606ANN-MM			
	160416R-MM	080608-MF			
	1604PDSR-MF	080608-MM			
	1604PDSR-MM	080608-W			
	LNMX	151008PNR-MM		0806ANN-MF	
				0806ANN-MM	
			ONMX	0606ANN-MF	
				0606ANN-MM	

* For more information, please refer KORLOY 2010-2011 catalogue

 PC9530(PVD)

Geometry	Designation	Geometry	Designation
	ONMX 080608-MF		SPMT 050203-DF
	080608-MM		050203-DM
	0806ANN-MF		050203-DS
	0806ANN-MM		060204-DF
	RDHW 0501M0E		060204-DM
	06T1M0E		060204-DS
	0702M0E		070204-DF
	0803M0E		070204-DM
	0803M0T		070204-DS
	RDKT 10T3M0-MF		TECN 32TR-S20
	10T3M0-MM		TEEN 43TR-S20
	1204M0-MF		43TR-ZH
	1204M0-MM		TPCN 1603PDR-S20
	SDXT 09M405R-MF		2204PDR-S20
	09M405R-MM		WDKT 080316ZDSR-MH
	130508R-MF		10T320ZDSR-MH
	130508R-MM		130520ZDSR-MH
	SECA 1204AFTN		150625ZDSR-MH
	1203AFTN-S20		WNMX 09T316ZNN-MM
	SNMX 1206ANN-MM		130520ZNN-MM

* For more information, please refer KORLOY 2010-2011 catalogue

 PC3545(PVD)

Geometry	Designation	Geometry	Designation
	ADKT 1505PDSR-MM		APMT 090312R-MM
			090316R-MM
	APKT 1604PDSR		090320R-MM
	1604PDSR-MM		11T3PDSR-MM
			11T3PDSR-MF
	APMT 060202PDSR-MM		11T308PDSR-MM
	0602PDSR-MM		11T312PDSR-MM
	060208PDSR-MM		11T316R-MM
	060212R-MM		11T324R-MM
	060216R-MM		1604PDSR-MM
	0903PDSR-MM		1604PDSR-MF
	090308PDSR-MM		160410PDSR-MM
			160416PDSR-MM

* For more information, please refer KORLOY 2010-2011 catalogue

Hard-to-cut material
cutting

New grade for Hard-
to-cut material

Technical guide
for Turning

Technical guide
for Milling

Technical guide
for Endmill

Workpiece
code table

Hard-to-cut material
cutting solution

Trouble shooting

PC3545(PVD)

Geometry	Designation	Geometry	Designation
	APMT 160424R-MM		SEKN 1504AFSN-SU
	160432R-MM		SEXN 1203AFSN-FM
	1806PDSR-MM		1504AFSN-FM
	1806PDSR-MF		SEXR 1203AFSN-FM
	1806PDSR-ML		SNCF 1206ANN-MM
	180612PDSR-MM		1206ENN-MM
	180612PDSR-MM		1206QNN-MM
	180616PDSR-MM		1507ANN-MM
	180620PDSR-MM		1507ENN-MM
	180624PDSR-MM		SPKN 1203EDSR-SM
	APXT 1035PDSR-MM		1504EDSR-SM
	11T3PDSR-MF		1504EDSR-SU
	11T3PDSR-MM		SPXN 1203EDSR-FM
	1604PDSR-MF		1504EDSR-FM
	1604PDSR-MM		SPXR 1203EDSR-FM
	LNEX 100605PNR-MM		1504EDSR-FM
	151008PNR-MM		TPKN 1603PDSR-SM
	LNMX 100605PNR-MM		1603PDSR-SU
	151008PNR-MM		2204PDSR-SM
	OFCN 0704SN		2204PDSR-SU
	OFKR 0704SN-MM		TPXN 1603PDSR-FM
			2204PDSR-FM
	RCMX 1003M0		TPXR 1603PDSR-FM
	RDKT 10T3M0-MF		2204PDSR-FM
	10T3M0-MM		WDKT 080316ZDSR-MH
	1204M0-MF		10T320ZDSR-MH
	1204M0-MM		130520ZDSR-MF
	1605M0-MM		150625ZDSR-MH
	2006M0-MM		WNMX 09T316ZNN-MM
	SDKN 1203AESN-SM		130520ZNN-MM
	1203AESN-SU		
	1504AESN-SM		
	1504AESN-SU		
	SDXN 1203AESN-FM		
	1504AESN-FM		
	SDXT 130508R-MM		
	SEKN 1203AFSN-SM		
	1203AFSN-SU		
	1504AFSN-SM		

* For more information, please refer KORLOY 2010-2011 catalogue

NC5330(CVD)

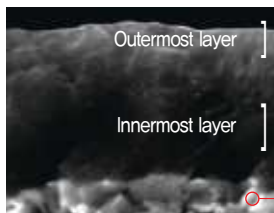
Geometry	Designation	Geometry	Designation
	ONHX 0606ANN-MM		14M4AGSN-MM
	0806ANN-MM		14M4AGTN-MF
	ONMX 0606ANN-MM		SEXT 0903AGSN-MF
	0806ANN-MM		0903AGSN-MM
	RDKT 10T3M0-MM		14M4AGSN-MF
	1204M0-MM		14M4AGSN-MM
			14M4AGSN-MR
	SDKR 1203AEN-MX		SNEX 1206ANN-MM
	1504AEN-MX		1206ENN-MM
	SECN 1203AFSN		SNMX 1206ANN-MM
	1504AFSN		1206ENN-MM
	SEKR 1203AFSN-MX		SPMT 120408
	1504AFSN-MX		
	SPCN 1203EDR		
	1504EDR		TEEN 43TR
	SPKN 1203EDSR		TPCN 1603PDR
	SPKR 1203EDSR-MX		2204PDR
	SDXT 09M405R-MF		TPKN 2204PDSR
	09M405R-MM		TPKR 2204PDSR-MX
	130508R-MF		WCMT 040208-C20
	130508R-MM		050308-C20
	SEET 0903AGSN-MF		06T308-C20
	0903AGSN-MM		080408-C20
	14M4AGSN-MF		

* For more information, please refer KORLOY 2010-2011 catalogue



Technical guide for Endmill

PC210 coating layer

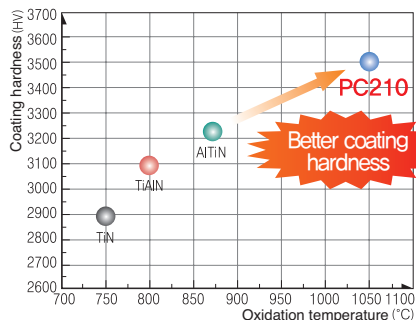


Outermost layer → Higher hardness & Anti oxidation

Innermost layer → Good bonding strength of coating and substrate Anti chipping

→ Ultra fine grain tough substrate

Coating hardness & Anti oxidation



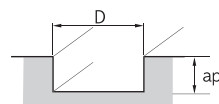
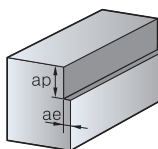
Features

- High rake angel and curvilinear chip pocket.
- Strong cutting edge against work hardening.
- Suitable for stainless steel, Ti-Alloy & Inconel.
- Shouldering, slotting & ramping.



Cutting Condition

Workpiece Condition	Stainless steel		Ti-alloy / Inconel	
	RPM n(min ⁻¹)	feed v _f (mm/min)	RPM n(min ⁻¹)	feed v _f (mm/min)
Condition(mm)				
2	5,500	240	2,600	90
4	4,000	260	2,000	90
6	3,000	360	1,200	90
8	2,000	390	1,000	100
10	1,700	410	800	120
12	1,500	380	700	100
14	1,200	320	600	95
16	1,000	270	500	90
20	750	250	400	85



- Shouldering
 - Steel, alloy steel, stainless steel : $ae=0.1D$, $ap=1.5D$
 - Ti-alloy, Inconel, HSS : $ae=0.05D$, $ap=1.5D$
- Slotting
 - Steel, alloy steel : $ap=1.0D$
 - Stainless steel : $ap=0.3D$
 - Ti-alloy, Inconel, HSS : $ap=0.2D$

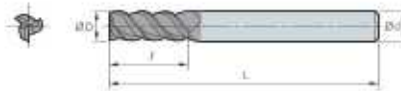
* For more information, please refer KORLOY 2010-2011 catalogue

Endmill stock management

PC210(PVD)Solid endmill



ØD	Tolerance
Ø1 ~ Ø 6	-0.01 ~ -0.030
Ø7 ~ Ø 10	-0.015 ~ -0.040
Ø11 ~ Ø 20	-0.020 ~ -0.050



(mm)

No.	ØD	Ød	ℓ	L	
IFSE	3030-045	3.0	6	10	45
	3035-045	3.5	6	10	45
	3040-045	4.0	6	12	45
	3045-045	4.5	6	12	45
	3050-050	5.0	6	15	50
	3055-050	5.5	6	15	50
	3060-050	6.0	6	15	50
	3065-060	6.5	8	20	60
	3070-060	7.0	8	20	60
	3075-060	7.5	8	20	60
	3080-060	8.0	8	20	60
	3085-070	8.5	10	20	70
	3090-070	9.0	10	20	70
	3095-070	9.5	10	20	70
	3100-070	10	10	25	70
	3110-075	11	12	25	75
	3120-075	12	12	30	75
	3130-090	13	16	30	90
	3140-090	14	16	35	90
	3150-090	15	16	40	90
	3160-090	16	16	40	90
	3170-100	17	20	40	100
	3180-100	18	20	45	100
	3190-110	19	20	45	110
	3200-110	20	20	45	110

* IFSE3000-L(000)-I(V00)

Ex1) 3 flutes, diameter : 6.3, l : 17, L : 60 => IFSE3063-060-V17

Workpiece code table

Stainless steel

Depth of	KOR	ISO	JPN	USA		ENG	GER	FRN	RUS	
	KS	ISO	JIS	UNS	AISI SAE	BS BS/EN	DIN DIN/EN	NF NF/EN	GOST	
Stainless steels	Austenite	STS201	X12CrMnNiN17-7-5	SUS201	S20100	201	284S16	X12CrNi17-7	Z12CMN17-07Az	12X17-9AH4
		STS202	X12CrMnNiN18-9-5	SUS202	S20200	202	301S21	X2CrNiN18-7		07X16H6
		STS301	X10CrNi18-8	SUS301	S30100	301		X12CrNi17-7	Z11CN17-08	
		STS301L	X2CrNiN18-7	SUS301L						
		STS301J1		SUS301J1			302S25			12X18H9
		STS302		SUS302	S30200	302		X10CrNiS18-9	Z12CN18-09	
		STS302B	X12CrNiSi18-9-3	SUS302B	S30215	302B	303S21			
		STS303	X10CrNiS18-9	SUS303	S30300	303	303S41		Z8CNF18-09	12X18H10E
		STS303Se		SUS303Se	S30323	303Se		X5CrNi18-10		
		STS303Cu		SUS303Cu			304S31			08X18H10
		STS304	X5CrNi18-9	SUS304	S30400	304		X2CrNi19-11	Z7CN18-09	
			X2CrNi18-9				304S11			03X18H11
		STS304L	X2CrNi19-11	SUS304L	S30403	304L		X2CrNiN18-10	Z3CN19-11	
		STS304N1	X5CrNiN18-8	SUS304N1	S30451	304N			Z6CN19-09Az	
		STS304LN	X2CrNiN18-8	SUS304LN	S30453	304LN		X5CrNi18-12	Z3CN18-10Az	
		STS304J1		SUS304J1			305S19			06X18H11
		STS305	X6CrNi18-12	SUS305	S30500	305			Z8CN18-12	
		STS309S		SUS309S	S30908	309S	310S31	X5CrNiMo27-12-2	Z10CN24-13	10X23H18
		STS310S	X6CrNi25-20	SUS310S	S31008	310S	316S31	X5CrNiMo27-13-3	Z8CN25-20	
		SUS316	X5CrNiMo17-12-2	SUS316	S31600	316		X2CrNiMo17-13-2	Z7CND17-12-02	
		X3CrNiMo17-12-3				316S11	X2CrNiMo17-14-3	Z6CND18-12-03	03X17H14M3	
	STS316L	X2CrNiMo17-12-2	SUS316L	S31603	316L			Z3CND17-12-02		
		X2CrNiMo17-12-3						Z3CND17-12-03		
		X2CrNiMo18-14-3								
	STS316N		SUS316N	S31651	316N	317S16	X6CrNiTi18-10			
	STS317		SUS317	S31700	317	321S31	X6CrNiNb18-10		08X18H10T	
	STS321	X6CrNiTi18-10	SUS321	S32100	321	347S31		Z6CNT18-10	08X18H12	
	STS347	X6CrNiNb18-10	SUS347	S34700	347		X6CrAl13	Z6CNNb18-10		
	STS384	X3NiCr18-16	SUS384	S38400	384	405S17		Z6CN18-16		
	Ferrite	STS405	X6CrAl13	SUS405	S40500	405			Z8CA12	
STS410L			SUS410L				X6Cr17	Z3C14		
STS429			SUS429	S42900	429	430S17	X7CrS18		12X17	
STS430		X6Cr17	SUS430	S43000	430		X6CrMo17-1	Z8C17		
STS430F		X7CrS17	SUS430F	S43020	430F	434S17		Z8CF17		
STS434		X6CrMo17-1	SUS434	S43400	434			Z8CD17-01		
STS444		X2CrMoTi18-2	SUS444	S44400	444			Z3CDT18-02		
STSM27			SUSXM27	S44627			X10Cr13	Z1CD26-01		
STS403			SUS403	S40300	403	410S21				
STS410		X12Cr13	SUS410	S41000	410	416S21	X20Cr13	Z13C13		
Martensite	STS416	X12CrS13	SUS416	S41600	416	420S29	X20CrNi17-2	Z11CF13	20X13	
	STS420J1	X20Cr13	SUS420J1	S42000	420	431S29		Z20C13	20X17H2	
	STS431	X19CrNi16-2	SUS431	S43100	431			Z15CN16-02		
	STS440A	X70CrMo15	SUS440A	S44002	440A		X7CrNiAl17-7	Z70C15		
	STS630	X5CrNiCuNb16-4	SUS630	S17400	S17400			Z6CNU17-04	09X17H7IO	
	STS631	X7CrNiAl17-7	SUS631	S17700	S17700			Z9CNA17-07		
Precipitation hardening	STS631J1		SUS631J1							



Heat resistant super alloy

Depth of		KOR	ISO	JPN	USA		ENG	GER	FRN	RUS	Commercial designation
		KS	ISO	JIS	UNS	AISI SAE	BS BS/EN	DIN DIN/EN	NF NF/EN	GOST	
Heat resistant alloy	Titanium alloy	-	-	-	AMS R54520	-	-	TiAl5Sn2.5	T-A5E	-	TA14/17
		-	-	-	AMS R56400	-	-	TiAl6V4	T-A6V	-	TA10-13
		-	-	-	AMS R56401	-	-	TiAl6V4ELI	-	-	TA11
	Nickel alloy	-	-	-	N10001	-	-	-	-	-	Hastelloy B
		-	-	-	N10276	-	-	-	-	-	Hastelloy C-276
		-	-	-	N06600	-	-	-	-	-	Inconel600
		-	-	-	N06625	-	-	-	-	-	Inconel625
		-	-	-	N09706	-	-	-	-	-	Inconel706
		-	-	-	N07718	-	-	-	-	-	Inconel718
		-	-	-	N09901	-	-	-	-	-	Nimonic901
	Cobalt alloy	-	-	-	R30006	-	-	-	-	-	Stellite 6
		-	-	-	R30021	-	-	-	-	-	Stellite 21
		-	-	-	R30031	-	-	-	-	-	Stellite 31
		-	-	-	-	-	-	-	-	-	-
	alloy steel	-	-	-	N08800	-	-	-	-	-	Incoloy 800
		-	-	-	S17400	-	-	-	-	-	17-4-PH
		-	-	-	-	-	-	-	-	-	-

Hard-to-cut material cutting

New grade for Hard-to-cut material

Technical guide for Turning

Technical guide for Milling

Technical guide for Endmill

Workpieces code table

Hard-to-cut material cutting solution

Trouble shooting

Hard-to-cut material cutting solution

Basic of stainless steel

Classification	Characteristic
Austenite	• General stainless steel with Cr & Ni, Food, chemical, construction.
Ferrite	• Stainless steel without Ni. Workability is quite good.
Martensite	• Heat treatable stainless steel. Higher C, Less corrosion resistance.
Precipitation hardening	• Cr–Ni base alloy. Good corrosion resistance and toughness
Austenite–Ferrite	• Good heat resistance Heat exchanger

Workability of stainless

- Work hardening : poor chip control, fast wear.
- Low thermal conduction : plastic deformation, fast wear.
- Built-up edge : micro chipping, poor surface roughness.
- Chemical affinity : chipping & breakage.

Solution for stainless cutting

- Try tool with good thermal conduction
To decrease heat concentration, use coolant or tool with good thermal conduction.
- Try shaper cutting edge
To decrease cutting load and prevent B.U.E, sharp edge and rake is essential.
- Apply proper cutting condition
Proper cutting speed to avoid work hardening and B.U.E should be applied.
- Try proper grade
Tough and high bonding substrate is required.

Trouble shooting

Trouble	Trouble shooting
Poor chip control (Turning)	• 1st choice : HS chip breaker • Too low depth of cut is not suitable due to work hardening ($a_p=0.5\sim 1\text{mm}$) • In case of low depth, HA chip breaker is recommended
B.U.E. (Built-up edge)	• Increase feed and cutting speed • In case of 'Ti-alloy' 'HA' is recommended (GS-HS-HA) • Higher rake tool is suitable.
Notch wear chipping	• 1st choice : 'GS' chip breaker • In case of breakage, 'VM' chip breaker is recommended
Breakage	• Check grade choice is suitable. • Please refer page 10. (Recommended cutting speed)
PVD vsCVD	• $a_p > 2\text{mm}$: CVD (NC9025) • $a_p < 2\text{mm}$: PVD (PC5300)
Poor wear resistance	• If it is continuous, Turning : PC8110 & NC9025 Milling : PC5300
Poor toughness	• For the better toughness than PC8110, try PC5300.
Welding part Breakage	• Grade : NC9025 & PC5300 • Chip breaker : GS, VM
Stainless steel Heat resistant super alloy General steel (Turning grade)	• Try NC9025 for various work material
Chipping by chip impact	• Down milling can be a solution • Change chip breaker with smooth chip flow
Breakage by intermittent cutting	• Try tougher grade
Coolant use	• High cutting speed with coolant can make work hardening due to rapid cooling. Therefore, decreasing cutting speed is recommended. (In case of intermittent, dry cutting is recommended.)

Hard-to-cut material
cutting

New grade for Hard-
to-cut material

Technical guide
for Turning

Technical guide
for Milling

Technical guide
for Endmill

Workpiece
code table

Hard-to-cut material
cutting solution

Trouble shooting

Memo





KORLOY Global Network

WE CREATE YOUR TOMORROW!

We stand to be the Global Top Manufacturer. Korea s Best. World s Best.

KORLOY stands to become the global company. With the beginning of the 21st century, along with Korloy s innovative settling of the global system and policy, we head to become the Korea s Best, World s Best, as the top manufacturer worldwide. KORLOY, as a cutting tool pioneer, is lined up with an ever-growing exporting network through the Overseas Sales Team to over 80 countries. Korloy commits to providing the best support and service worldwide.



This catalogue is made current on February 2010 as per ISO standard. The information contained in this catalogue, such as dimensions, perspective performance and quality, is subject to change due to advancing research and development.

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